Danko, Bence - assignment5

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***Abstract*—Backpropogation derivatives for various model architectures.**

***Index Terms*—Calculus**

I. Question 1

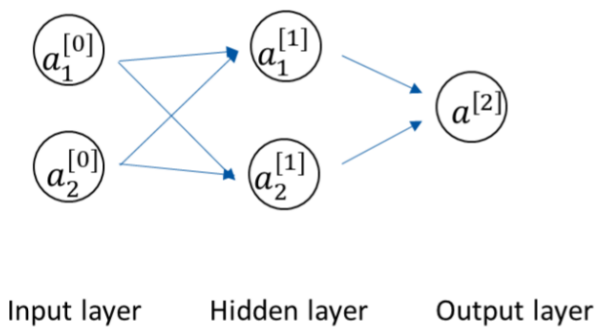

Fig. 1. Neural Network for Question 1. Bias terms in the hidden layer and output layer, sigmoid activation functions for both layers.

A. Forward Pass

1) Hidden Layer:

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

2) Output layer:

$$Z^{[2]} = W_0^{[2]} + W_1^{[2]} a_1^{[1]} + W_2^{[2]} a_2^{[1]}$$

$$a^{[2]} = \sigma(Z^{[2]})$$

B. Backpropagation

1) Binary Cross Entropy Loss Derivative: We assume binary cross entropy loss:

$$L(a^{[2]}, y) = -[y \cdot \log(a^{[2]}) + (1-y) \cdot \log(1 - a^{[2]})]$$

$$\frac{d}{dx} \log(x) = \frac{1}{x}$$

$$\frac{d}{dx} \log(1-x) = -\frac{1}{1-x}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial a^{[2]}} = -\left[y \cdot \frac{1}{a^{[2]}} + (1-y) \cdot \left(-\frac{1}{1-a^{[2]}} \right) \right]$$

$$\frac{\partial L(a^{[2]}, y)}{\partial a^{[2]}} = -\frac{y}{a^{[2]}} + \frac{1-y}{1-a^{[2]}}$$

2) Derivative of Sigmoid Activation:

$$\frac{da}{dz} = \frac{d}{dz}(1+e^{-z})^{-1}$$

$$\frac{da}{dz} = -1(1+e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$\frac{da}{dz} = \left(\frac{1}{1+e^{-z}} \right) \cdot \left(\frac{1+e^{-z}-1}{1+e^{-z}} \right)$$

$$\frac{da}{dz} = \sigma(z) \cdot (1 - \sigma(z))$$

C. Output Layer Weights

We calculate the gradient for each weight by multiplying the loss derivative, the activation derivative, and the logit derivative.

1) Weight $W_0^{[2]}$:

$$\frac{\partial L(a^{[2]}, y)}{\partial W_0^{[2]}} = \left(\frac{\partial L(a^{[2]}, y)}{\partial a^{[2]}} \right) \cdot \left(\frac{\partial a^{[2]}}{\partial Z^{[2]}} \right) \cdot \left(\frac{\partial Z^{[2]}}{\partial W_0^{[2]}} \right)$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_0^{[2]}} = \left(-\frac{y}{a^{[2]}} + \frac{1-y}{1-a^{[2]}} \right) \cdot \left(a^{[2]}(1-a^{[2]}) \right) \cdot (1)$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_0^{[2]}} = a^{[2]} - y$$

2) Weight $W_1^{[2]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial W_1^{[2]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a^{[2]}}\right)\cdot\left(\frac{\partial a^{[2]}}{\partial Z^{[2]}}\right)\cdot\left(\frac{\partial Z^{[2]}}{\partial W_1^{[2]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_1^{[2]}} = \left(-\frac{y}{a^{[2]}}+\frac{1-y}{1-a^{[2]}}\right)\cdot\left(a^{[2]}(1-a^{[2]})\right)\cdot(a_1^{[1]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_1^{[2]}} = (a^{[2]}-y)a_1^{[1]}$$

3) Weight $W_2^{[2]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial W_2^{[2]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a^{[2]}}\right)\cdot\left(\frac{\partial a^{[2]}}{\partial Z^{[2]}}\right)\cdot\left(\frac{\partial Z^{[2]}}{\partial W_2^{[2]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_2^{[2]}} = \left(-\frac{y}{a^{[2]}}+\frac{1-y}{1-a^{[2]}}\right)\cdot\left(a^{[2]}(1-a^{[2]})\right)\cdot(a_2^{[1]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_2^{[2]}} = (a^{[2]}-y)a_2^{[1]}$$

D. Hidden Layer

For the hidden layer, we must first find how the loss changes with respect to the hidden activations $a_i^{[1]}$.

1) Hidden Node 1 Weights: First, we find the gradient of the loss with respect to the activation $a_1^{[1]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a^{[2]}}\right)\cdot\left(\frac{\partial a^{[2]}}{\partial Z^{[2]}}\right)\cdot\left(\frac{\partial Z^{[2]}}{\partial a_1^{[1]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}} = \left(-\frac{y}{a^{[2]}}+\frac{1-y}{1-a^{[2]}}\right)\cdot\left(a^{[2]}(1-a^{[2]})\right)\cdot(W_1^{[2]})$$

Now we apply the chain rule for the weights in Node 1:

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}}\right)\cdot\left(\frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}\right)\cdot\left(\frac{\partial Z_1^{[1]}}{\partial W_{11}^{[1]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}}\right)\cdot\left(a_1^{[1]}(1-a_1^{[1]})\right)\cdot(a_1^{[0]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}} = (a^{[2]}-y)W_1^{[2]}\cdot a_1^{[1]}(1-a_1^{[1]})\cdot a_1^{[0]}$$

2) Hidden Node 2: First, we find the gradient of the loss with respect to the activation $a_2^{[1]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a^{[2]}}\right)\cdot\left(\frac{\partial a^{[2]}}{\partial Z^{[2]}}\right)\cdot\left(\frac{\partial Z^{[2]}}{\partial a_2^{[1]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}} = \left(-\frac{y}{a^{[2]}}+\frac{1-y}{1-a^{[2]}}\right)\cdot\left(a^{[2]}(1-a^{[2]})\right)\cdot(W_2^{[2]})$$

Now we apply the chain rule for the weights in Node 2:

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}}\right)\cdot\left(\frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}}\right)\cdot\left(\frac{\partial Z_2^{[1]}}{\partial W_{21}^{[1]}}\right)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}}\right)\cdot\left(a_2^{[1]}(1-a_2^{[1]})\right)\cdot(a_1^{[0]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}} = (a^{[2]}-y)W_2^{[2]}\cdot a_2^{[1]}(1-a_2^{[1]})\cdot a_1^{[0]}$$

II. Final Weight Derivations

$$\frac{\partial L(a^{[2]},y)}{\partial W_0^{[2]}} = a^{[2]}-y$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_1^{[2]}} = (a^{[2]}-y)a_1^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_2^{[2]}} = (a^{[2]}-y)a_2^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[1]}} = (a^{[2]}-y)W_1^{[2]}\cdot a_1^{[1]}(1-a_1^{[1]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}} = (a^{[2]}-y)W_1^{[2]}\cdot a_1^{[1]}(1-a_1^{[1]})\cdot a_1^{[0]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[1]}} = (a^{[2]}-y)W_1^{[2]}\cdot a_1^{[1]}(1-a_1^{[1]})\cdot a_2^{[0]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[1]}} = (a^{[2]}-y)W_2^{[2]}\cdot a_2^{[1]}(1-a_2^{[1]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}} = (a^{[2]}-y)W_2^{[2]}\cdot a_2^{[1]}(1-a_2^{[1]})\cdot a_1^{[0]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[1]}} = (a^{[2]}-y)W_2^{[2]}\cdot a_2^{[1]}(1-a_2^{[1]})\cdot a_2^{[0]}$$

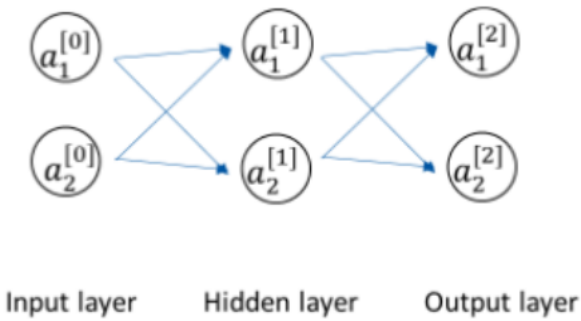

Fig. 2. Neural Network for Question 2. Bias terms in the hidden layer and output layer. We use sigmoid activation for the first layer, and softmax activation for the output layer.

III. Question 2

A. Forward Pass

1) Hidden Layer:

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

2) Output layer:

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]}$$

$$a_1^{[2]} = \frac{\exp(Z_1^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]})}$$

$$a_2^{[2]} = \frac{\exp(Z_2^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]})}$$

B. Backpropagation

1) Categorical Cross Entropy Loss Derivative: We assume categorical cross entropy loss for the multiple output classes:

$$L(a^{[2]}, y) = -\sum_{k=1}^{2} y_k \cdot \log(a_k^{[2]})$$

$$\frac{\partial L(a^{[2]}, y)}{\partial a_k^{[2]}} = -\frac{y_k}{a_k^{[2]}}$$

2) Derivative of Softmax Activation:

$$a_i^{[2]} = \frac{\exp(Z_1^{[2]})}{\sum_{j=1}^{K} \exp(Z_j^{[2]})}$$

When $i = j$: $\frac{\partial a_i^{[2]}}{\partial Z_i^{[2]}} = a_i^{[2]}(1 - a_i^{[2]})$

When $i \neq j$: $\frac{\partial a_i^{[2]}}{\partial Z_j^{[2]}} = -a_i^{[2]} a_j^{[2]}$

Applying the chain rule to find the gradient of the loss with respect to the logit $Z_i^{[2]}$:

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = \sum_{k=1}^{2} \left(\frac{\partial L(a^{[2]}, y)}{\partial a_k^{[2]}} \right) \cdot \left(\frac{\partial a_k^{[2]}}{\partial Z_i^{[2]}} \right)$$

For a specific node i (and letting the other node be j):

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = \left(-\frac{y_i}{a_i^{[2]}} \right) \cdot a_i^{[2]}(1 - a_i^{[2]}) + \left(-\frac{y_j}{a_j^{[2]}} \right) \cdot (-a_j^{[2]} a_i^{[2]})$$

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = -y_i(1 - a_i^{[2]}) + y_j a_i^{[2]}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = -y_i + y_i a_i^{[2]} + y_j a_i^{[2]}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = a_i^{[2]}(y_i + y_j) - y_i$$

We let $y_i + y_j = 1$:

$$\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}} = a_i^{[2]} - y_i$$

C. Output Layer Weights

We calculate the gradient for each weight by multiplying the loss derivative with respect to the logit ($\frac{\partial L(a^{[2]}, y)}{\partial Z_i^{[2]}}$) and the logit derivative with respect to the weight.

1) Weights for Output Node 1:

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{10}^{[2]}} = \left(\frac{\partial L(a^{[2]}, y)}{\partial Z_1^{[2]}} \right) \cdot \left(\frac{\partial Z_1^{[2]}}{\partial W_{10}^{[2]}} \right) = (a_1^{[2]} - y_1) \cdot (1) = a_1^{[2]} - y_1$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{11}^{[2]}} = \left(\frac{\partial L(a^{[2]}, y)}{\partial Z_1^{[2]}} \right) \cdot \left(\frac{\partial Z_1^{[2]}}{\partial W_{11}^{[2]}} \right) = (a_1^{[2]} - y_1) \cdot (a_1^{[1]})$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{12}^{[2]}} = \left(\frac{\partial L(a^{[2]}, y)}{\partial Z_1^{[2]}} \right) \cdot \left(\frac{\partial Z_1^{[2]}}{\partial W_{12}^{[2]}} \right) = (a_1^{[2]} - y_1) \cdot (a_2^{[1]})$$

2) Weights for Output Node 2:

$$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[2]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial Z_2^{[2]}}\right)\cdot\left(\frac{\partial Z_2^{[2]}}{\partial W_{20}^{[2]}}\right) = (a_2^{[2]}-y_2)\cdot(1) = a_2^{[2]}-y_2$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[2]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial Z_2^{[2]}}\right)\cdot\left(\frac{\partial Z_2^{[2]}}{\partial W_{21}^{[2]}}\right) = (a_2^{[2]}-y_2)\cdot(a_1^{[1]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[2]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial Z_2^{[2]}}\right)\cdot\left(\frac{\partial Z_2^{[2]}}{\partial W_{22}^{[2]}}\right) = (a_2^{[2]}-y_2)\cdot(a_2^{[1]})$$

D. Hidden Layer

For the hidden layer, we must first find how the loss changes with respect to the hidden activations $a_i^{[1]}$. Because each hidden node connects to multiple output nodes, we sum the gradients propagating backwards from all paths.

1) Hidden Node 1 Weights: First, we find the gradient of the loss with respect to the activation $a_1^{[1]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial Z_1^{[2]}}\right)\cdot\left(\frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}}\right) + \left(\frac{\partial L(a^{[2]},y)}{\partial Z_2^{[2]}}\right)\cdot\left(\frac{\partial Z_2^{[2]}}{\partial a_1^{[1]}}\right) = (a_1^{[2]}-y_1)W_{11}^{[2]} + (a_2^{[2]}-y_2)W_{21}^{[2]}$$

Now we apply the chain rule for the weights in Hidden Node 1:

$$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}}\right)\cdot\left(\frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}\right)\cdot\left(\frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{11}^{[2]} + (a_2^{[2]}-y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1-a_1^{[1]})\cdot(1)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}}\right)\cdot\left(\frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}\right)\cdot\left(\frac{\partial Z_1^{[1]}}{\partial W_{11}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{11}^{[2]} + (a_2^{[2]}-y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1-a_1^{[1]})\cdot(a_1^{[0]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_1^{[1]}}\right)\cdot\left(\frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}\right)\cdot\left(\frac{\partial Z_1^{[1]}}{\partial W_{12}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{11}^{[2]} + (a_2^{[2]}-y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1-a_1^{[1]})\cdot(a_2^{[0]})$$

2) Hidden Node 2: First, we find the gradient of the loss with respect to the activation $a_2^{[1]}$:

$$\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial Z_1^{[2]}}\right)\cdot\left(\frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}}\right) + \left(\frac{\partial L(a^{[2]},y)}{\partial Z_2^{[2]}}\right)\cdot\left(\frac{\partial Z_2^{[2]}}{\partial a_2^{[1]}}\right) = (a_1^{[2]}-y_1)W_{12}^{[2]} + (a_2^{[2]}-y_2)W_{22}^{[2]}$$

Now we apply the chain rule for the weights in Hidden Node 2:

$$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}}\right)\cdot\left(\frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}}\right)\cdot\left(\frac{\partial Z_2^{[1]}}{\partial W_{20}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{12}^{[2]} + (a_2^{[2]}-y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1-a_2^{[1]})\cdot(1)$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}}\right)\cdot\left(\frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}}\right)\cdot\left(\frac{\partial Z_2^{[1]}}{\partial W_{21}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{12}^{[2]} + (a_2^{[2]}-y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1-a_2^{[1]})\cdot(a_1^{[0]})$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[1]}} = \left(\frac{\partial L(a^{[2]},y)}{\partial a_2^{[1]}}\right)\cdot\left(\frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}}\right)\cdot\left(\frac{\partial Z_2^{[1]}}{\partial W_{22}^{[1]}}\right) = \left[(a_1^{[2]}-y_1)W_{12}^{[2]} + (a_2^{[2]}-y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1-a_2^{[1]})\cdot(a_2^{[0]})$$

IV. Final Weight Derivations

$$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[2]}} = a_1^{[2]}-y_1$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{11}^{[2]}} = (a_1^{[2]}-y_1)a_1^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{12}^{[2]}} = (a_1^{[2]}-y_1)a_2^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{20}^{[2]}} = a_2^{[2]}-y_2$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{21}^{[2]}} = (a_2^{[2]}-y_2)a_1^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{22}^{[2]}} = (a_2^{[2]}-y_2)a_2^{[1]}$$

$$\frac{\partial L(a^{[2]},y)}{\partial W_{10}^{[1]}} = \left[(a_1^{[2]}-y_1)W_{11}^{[2]} + (a_2^{[2]}-y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1-a_1^{[1]})$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{11}^{[1]}} = \left[(a_1^{[2]} - y_1)W_{11}^{[2]} + (a_2^{[2]} - y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1 - a_1^{[1]}) \cdot a_1^{[0]}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{12}^{[1]}} = \left[(a_1^{[2]} - y_1)W_{11}^{[2]} + (a_2^{[2]} - y_2)W_{21}^{[2]}\right] \cdot a_1^{[1]}(1 - a_1^{[1]}) \cdot a_2^{[0]}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{20}^{[1]}} = \left[(a_1^{[2]} - y_1)W_{12}^{[2]} + (a_2^{[2]} - y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1 - a_2^{[1]})$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{21}^{[1]}} = \left[(a_1^{[2]} - y_1)W_{12}^{[2]} + (a_2^{[2]} - y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1 - a_2^{[1]}) \cdot a_1^{[0]}$$

$$\frac{\partial L(a^{[2]}, y)}{\partial W_{22}^{[1]}} = \left[(a_1^{[2]} - y_1)W_{12}^{[2]} + (a_2^{[2]} - y_2)W_{22}^{[2]}\right] \cdot a_2^{[1]}(1 - a_2^{[1]}) \cdot a_2^{[0]}$$